Juan Andrés Betancourt Saco

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Education

Universidad del Norte — Barranquilla, Colombia
B.Sc. in Electronic Engineering — *Expected Sep. 2027*

- Coursework completed Dec. 2026. Saber Pro national exit exam: **99th percentile**.
- Relevant coursework: Microcontrollers, Telematics, Communications, Antennas and Radio Links, Cellular Systems, Automatic Control, Measurements and Instrumentation.

Projects

Early Corrosion Detection for Steel | *YOLOv8, ONNX, AWS CDK, Lambda, DynamoDB, React* — 2026

- Led the technical design of an industrial system that detects early corrosion on steel plates from photographs taken under ASTM B117 salt spray testing.
- Built a serverless AWS backend in CDK (Python): four stacks with S3, DynamoDB, Cognito, API Gateway and six Lambda functions, including YOLOv8 segmentation inference on ONNX Runtime; set up model training on about 32,000 images and built a React dashboard with a plant map, image overlays and alerts.

Classroom Occupancy with Edge AI | *Raspberry Pi 4, YOLOv8, LTE, AWS IoT Core, Next.js* — 2025

- Person detection on a Raspberry Pi 4 sending events over an LTE modem (SIMCOM A7670SA) to AWS IoT Core, Lambda and DynamoDB, shown on a Next.js dashboard and a Potree 3D point cloud.
- Raised on-device throughput from 2–3 to 6–10 FPS by tuning resolution, inference size and frame skipping; event-driven uploads reduced mobile data use.

Aria, Glove-Teleoperated Robotic Hand | *ESP32-S3, ESP-NOW, PCA9685, I^2C, C++* — 2026

- Two-board wireless system: an ESP32-S3 glove with five IR flex sensors drives five servos on an ESP32 hand through a PCA9685, over an ESP-NOW link with 1–10 ms latency.

UrbanTrack, GPS Vehicle Tracking | *AWS EC2, RDS PostgreSQL, VPC Peering, Node.js* — 2026

- GPS tracking web app with live and historical routes; writer and reader services on EC2 with PostgreSQL on RDS over VPC peering.

LTE Network Design, Bogotá (in progress) | *Link budget, COST-231 Hata, RF planning, Python, KML* — 2026

- Coverage dimensioning for 47.68 km^2 in band 2 (1900 MHz FDD): uplink-limited cell radius of 561 m, giving 78 three-sector eNBs on an 842 m hexagonal grid with 97.4% area coverage.

Pyramidal Horn Antenna for the Microwave Band | *Python, CST Microwave Studio* — 2026

- Python tool that sizes a horn antenna from frequency (1–20 GHz) and target gain (10–30 dB), validated in CST. Editor of the paper in preparation.

Experience

Kiddie Keep Well Camp — Edison, NJ, USA
Counselor, Head Counselor, Program Director, Assistant Director — *Summers 2023 – 2026*

- Promoted every season at a nonprofit residential camp for children from low-income families; as Program Director, designed the activity curriculum and led the counselor team.

IEEE Robotics and Automation Society, Student Chapter — Barranquilla, Colombia
President (2023 – 2024), Vice President (2022 – 2023) — *2022 – 2024*

- Grew the chapter from about 4 to 40 members through technical workshops, invited talks and student projects.

Technical Skills

Embedded: ESP32, ESP32-S3, Raspberry Pi 4, Arduino Core 3.x, SPI, I^2C, PWM, ESP-NOW, BLE, LTE modems
Cloud and IoT: AWS IoT Core, Lambda, DynamoDB, S3, Cognito, API Gateway, EC2, RDS, AWS CDK, MQTT
Programming: C, C++, Python, JavaScript, TypeScript, SQL; React, Next.js, Node.js
Vision and ML: YOLOv8, ONNX Runtime, OpenCV, MediaPipe, TensorFlow Lite
Hardware and Tools: KiCad, LTspice, CST Microwave Studio, MATLAB, Git, Linux, Docker
Languages: Spanish (native), English (B2), French (B1), Portuguese (A2), Turkish (A1)